

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
8	(a)	(i)		maximum possible mass = 47.2 (3) if incorrect credit following steps in reacting masses calculation 117 of NaCl gives 46 of Na (1) 120 of NaCl gives $\frac{46}{117} \times 120$ of Na (1) ecf possible accept alternative method using moles n(NaCl) = 2051 (1) n(Na) = 2051 (1) m = 2051 $\times$ 23 = 47173 m = 47.2 (1) percentage yield = 80.6 / 81 / 80.65 (1) ecf possible						
		(ii)		accept any of following for (1) - sodium chloride used was impure - not all the sodium chloride had reacted - side reactions taking place - loss of product	1			1		1
		(iii)		sodium would react with any water present		1		1		1

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	(b)	(i)		6.92 (3) award (2) for value given to any other number of sig figs $(6 \times 7.59) + (7 \times 92.41)$ (1) 6.9241 (1) ecf possible following minor slip	1	2		3	3	
		(ii)		the nucleus of lithium-7 contains four neutrons and that of lithium-6 contains three neutrons		1		1		
				Question 8 total	2	8	0	10	7	2